CSEC Add Maths

Paper 2

June 2025

Questions

SECTION I

ALGEBRA, SEQUENCES AND SERIES

ALL working must be clearly shown.

1. (a) Solve the equation $3^{2x+1} = 21$, giving your answer in exact form. [4]

$$3^{2x+1} = 21$$

$$3^{2x+1} = 7 \times 3^1$$

$$\frac{3^{2x+1}}{3^1} = 7$$

$$3^{2x+1-1} = 7$$

$$3^{2x} = 7$$

Taking ln on both sides:

$$\ln 3^{2x} = \ln 7$$

$$2x \ln 3 = \ln 7$$

$$2x = \frac{\ln 7}{\ln 3}$$

$$x = \frac{\ln 7}{2\ln 3}$$

(b) Determine the values of x and y which satisfy the equations

$$y + 3x = 2 \quad \text{and} \quad y + 4 = x(5 - 2x).$$ [6]

$y + 3x = 2$ —— Equation 1

$y + 4 = x(5 - 2x)$ —— Equation 2

Rearranging Equation 1 to make y the subject of the formula:

$y = 2 - 3x$ —— Equation 3

Substituting Equation 3 into Equation 2:

$$(2 - 3x) + 4 = x(5 - 2x)$$

$$2 - 3x + 4 = 5x - 2x^2$$

$$-3x + 6 = 5x - 2x^2$$

$$2x^2 - 5x - 3x + 6 = 0$$

$$2x^2 - 8x + 6 = 0$$

Dividing throughout by 2:

$$x^2 - 4x + 3 = 0$$

$$(x - 1)(x - 3) = 0$$

Either

$x - 1 = 0$ or $x - 3 = 0$

$x = 1$ or $x = 3$

When $x = 1$,

$y = 2 - 3(1)$

$y = 2 - 3$

$y = -1$

When $x = 3$,

$y = 2 - 3(3)$

$y = 2 - 9$

$y = -7$

$\therefore$ The values of x and y which satisfy the equations are:

$x = 1, y = -1$ and $x = 3, y = -7$.

(c) (i) Determine the solution set for the inequality $\frac{(3x+2)}{(x-1)} \leq 0$, giving your answer in set builder notation. [4]

$$\frac{(3x+2)}{(x-1)} \leq 0$$

Multiplying by $(x-1)^2$:

$$\frac{(3x+2)}{(x-1)} \times (x-1)^2 \leq 0 \times (x-1)^2$$

$$(3x+2)(x-1) \leq 0$$

Roots occur when

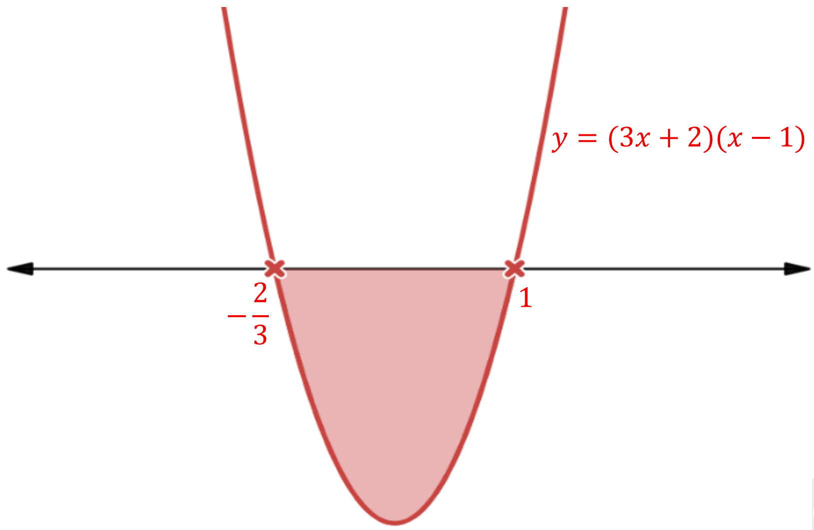
$3x + 2 = 0$ and $x - 1 = 0$

$3x = -2$ and $x = 1$

$$x = -\frac{2}{3}$$

- The coefficient of x^2 is positive therefore, this graph will have a <u>minimum curve</u>.
- Roots: $\left(-\frac{2}{3}, 0\right)$ and $(1, 0)$
- Since $(3x+2)(x-1) \leq 0$, this is the region where $y \leq 0$.

A sketch of the curve $y = (3x+2)(x-1)$ looks like:



$\therefore$ The solution set is $\{x \colon -\frac{2}{3} \leq x < 1\}$

(ii) State how the solution of $(3x+2)(x-1) \leq 0$ would differ from the solution of $\frac{(3x+2)}{(x-1)} \leq 0$ [1]

The solution of $(3x+2)(x-1) \leq 0$ is $\{x: -\frac{2}{3} \leq x \leq 1\}$.

The solution of $\frac{(3x+2)}{(x-1)} \leq 0$ is $\{x: -\frac{2}{3} \leq x < 1\}$.

The difference between the two solutions is the exclusion of 1 in the solution for $\frac{(3x+2)}{(x-1)} \leq 0$.

1 is excluded because it would make the function undefined.

Total: 15 marks

2. (a) (i) The relationship between two variables, x and y, is $10^{2x} \times 10^{y} = 3$. Express the relationship in the form $y = \log b + ax$, where $a, b \in R$. [4]

$10^{2x} \times 10^{y} = 3$

$10^{2x+y} = 3$

Taking log_{10} on both sides:

$log_{10}\,(10^{2x+y}) = log_{10}\,3$

$log_{10}\,10^{2x+y} = log_{10}\,3$

$(2x + y)log_{10} = log_{10}\,3$

$$2x + y = \frac{log_{10}\,3}{log_{10}\,10}$$

$$2x + y = \frac{log_{10}\,3}{1}$$

$2x + y = log_{10}\,3$

$y = log_{10}\,3 - 2x$

This is of the form $y = \log b + ax$ where $a = -2$ and $b = 3$.

(ii) Hence, state the value of the y-intercept of a graph of y versus x. [1]

$y = log_{10} 3 - 2x$

Rearranging:

$y = -2x + log_{10} 3$

This is of the form $y = mx + c$ where $m = -2$ and $c = log_{10} 3$.

Therefore, the value of the y-intercept of a graph is $c = log_{10} 3$.

(b) Determine whether the series $2 + 6 + 18 + 54 + 162 \ldots$ is arithmetic or geometric. Justify your response. [2]

An arithmetic series has a common difference between the terms.

From the series given, there is no common difference, d as shown below:

$T_2 - T_1$: $6 - 2 = 4$

$T_3 - T_2$: $18 - 6 = 12$

$T_4 - T_3$: $54 - 18 = 36$

$T_5 - T_4$: $162 - 54 = 108$

Hence, it is not an Arithmetic Progression.

A geometric series has a common ratio between the terms.

From the series given, there is a common ratio, r as shown below:

$\frac{T_2}{T_1}$: $\frac{6}{2} = 3$

$\frac{T_3}{T_2}$: $\frac{18}{6} = 3$

$\frac{T_4}{T_3}$: $\frac{54}{18} = 3$

$\frac{T_5}{T_4}$: $\frac{162}{54} = 3$

Since the sequence has a common ratio, it is geometric in nature.

(c) In a geometric progression, the 3rd term is 25 and the sum of the 1st and 2nd terms is 500. Find the common ratio where $r > 0$. [4]

In a geometric progression, $T_n = ar^{n-1}$, where $T_n = n^{th}$ term, $a = 1^{st}$ term, $r =$ common ratio.

From the question, the following information is given:

"The 3rd term is 25."

$$T_3 = 25$$

$$ar^2 = 25$$

$$a = \frac{25}{r^2} \quad \boxed{\text{Equation 1}}$$

"The sum of the 1st and 2nd terms is 500"

$$T_1 + T_2 = 500$$

$$a + ar = 500 \quad \boxed{\text{Equation 2}}$$

Substituting Equation 1 into Equation 2:

$$\left(\frac{25}{r^2}\right) + \left(\frac{25}{r^2}\right)r = 500$$

$$\frac{25}{r^2} + \frac{25}{r} = 500$$

$$\frac{25 + 25r}{r^2} = 500$$

$$25 + 25r = 500r^2$$

$$0 = 500r^2 - 25r - 25$$

$$500r^2 - 25r - 25 = 0$$

Dividing throughout by 25:

$$20r^2 - r - 1 = 0$$

$$(5r + 1)(4r - 1) = 0$$

Either

$5r + 1 = 0$ or $4r - 1 = 0$

$5r = -1$ or $4r = 1$

$r = -\frac{1}{5}$ or $r = \frac{1}{4}$

Since $r > 0$, then $r = \frac{1}{4}$ only.

(d) A kitchen garden has an area, A, $(10-3\sqrt{5})\ m^2$. The length of the garden is $(5-2\sqrt{5})\ m$. Determine the value of a, if the width of the garden, in metres, is in the form $(a+\sqrt{5})$, where a is a real number. [4]

Assuming the kitchen garden is rectangular, a sketch is shown below.

$Length = (5-2\sqrt{5})\ m$

$Area = (10-3\sqrt{5})\ m^2$

$Width = (a+\sqrt{5})\ m$

<u>Method 1</u>

$$Area = length \times width$$

$$Width = \frac{Area}{length}$$

$$= \frac{(10-3\sqrt{5})}{(5-2\sqrt{5})}$$

Rationalizing the denominator:

$$Width = \frac{(10-3\sqrt{5})}{(5-2\sqrt{5})} \times \frac{(5+2\sqrt{5})}{(5+2\sqrt{5})}$$

$$= \frac{50+20\sqrt{5}-15\sqrt{5}-6(5)}{25+10\sqrt{5}-10\sqrt{5}-4(5)}$$

$$= \frac{20 + 5\sqrt{5}}{5}$$

$$= \frac{5(4 + \sqrt{5})}{5}$$

$$= 4 + \sqrt{5}$$

which is of the form $(a + \sqrt{5})$

Therefore, $a = 4$.

Method 2

$$Area = length \times width$$

$$(10 - 3\sqrt{5}) = (5 - 2\sqrt{5}) \times (a + \sqrt{5})$$

$$10 - 3\sqrt{5} = 5a + 5\sqrt{5} - 2a\sqrt{5} - 2(5)$$

$$10 - 3\sqrt{5} = 5a - 10 + 5\sqrt{5} - 2a\sqrt{5}$$

$$10 - 3\sqrt{5} = (5a - 10) + (5 - 2a)\sqrt{5}$$

Equating coefficients:

$$10 = 5a - 10$$

$$10 + 10 = 5a$$

$$20 = 5a$$

$$\frac{20}{5} = a$$

$a = 4$

OR

$-3 = 5 - 2a$

$-3 - 5 = -2a$

$-8 = -2a$

$\frac{-8}{-2} = a$

$a = 4$

Therefore, $a = 4$.

Total: 15 marks

SECTION II

COORDINATE GEOMETRY, VECTORS AND TRIGONOMETRY

ALL working must be clearly shown.

3. (a) (i) Using the grid on page 11, draw the graphs of $\sin 2x$ and $\cos 2x$, for

$0 \leq x \leq \pi$ [2]

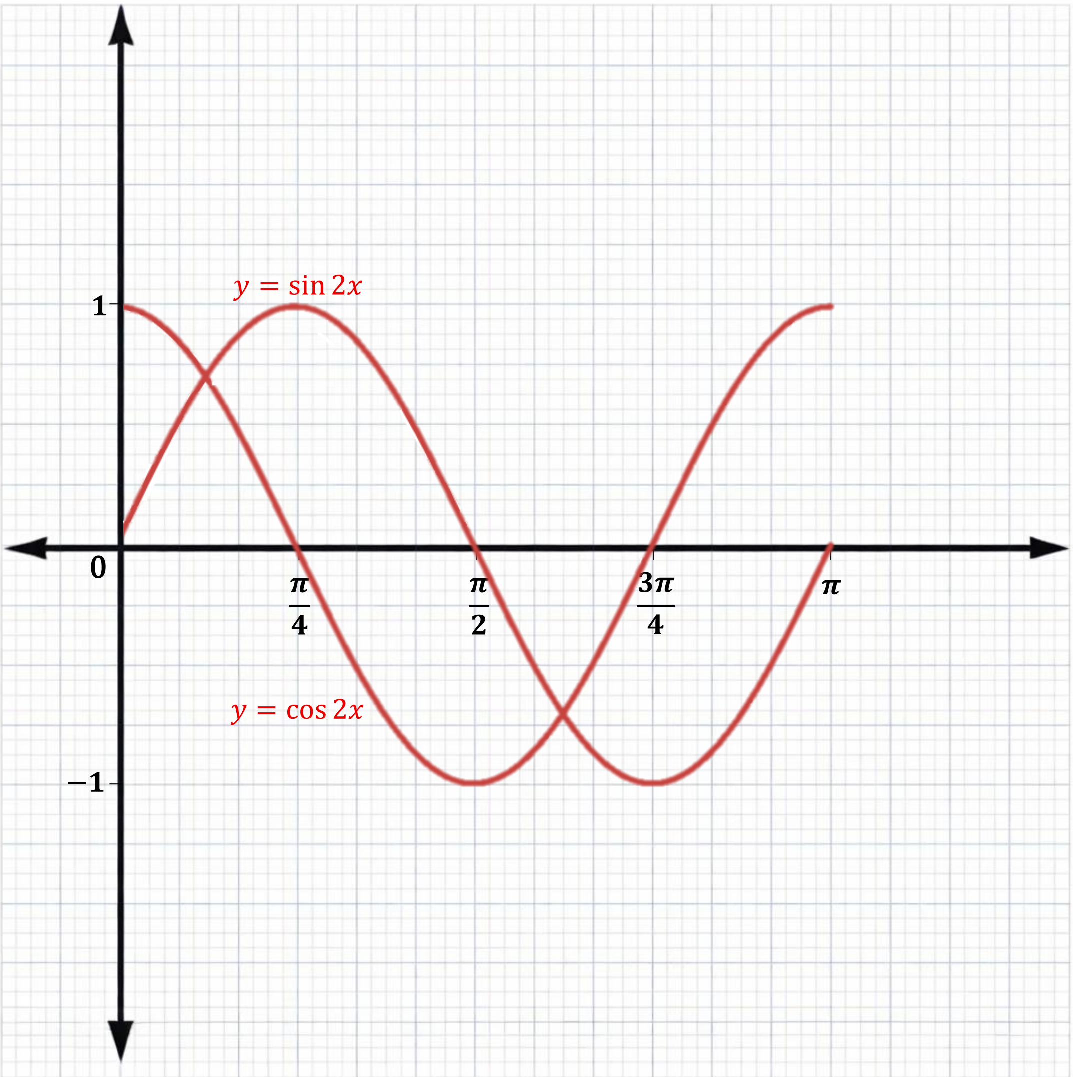
Table of values for $y = sin\ 2x$

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
$\sin 2x$	0	1	0	-1	0

Table of values for $y = cos\ 2x$

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
$\cos 2x$	1	0	-1	0	1

The graphs of both $\sin 2x$ and $\cos 2x$ can be seen on the next page.



$y = \sin 2x$
1
0
-1
$\frac{\pi}{4}$
$\frac{\pi}{2}$
$\frac{3\pi}{4}$
π
$y = \cos 2x$

(ii) Hence or otherwise, solve the equation $\sin 2x = \cos 2x$ for $0 \leq x \leq \pi$. [4]

. $\sin 2x = \cos 2x$

$$\frac{\sin 2x}{\cos 2x} = \frac{\cos 2x}{\cos 2x}$$

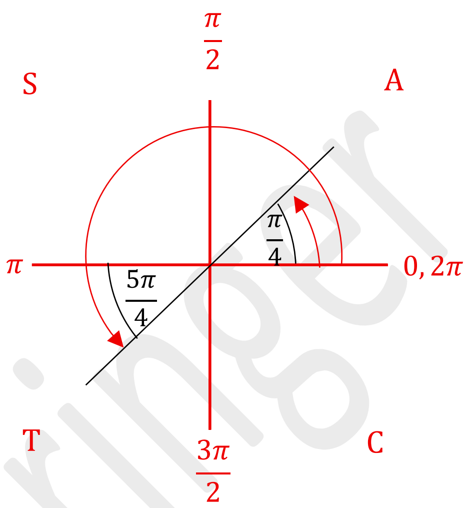
$\tan 2x = 1$

$$2x = \tan^{-1}(1)$$

$$2x = \frac{\pi}{4}, \left(\pi + \frac{\pi}{4}\right)$$

$$2x = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$x = \frac{\pi}{8}, \frac{5\pi}{8}$$



$\therefore x = \frac{\pi}{8}, \frac{5\pi}{8}$ for $0 \leq x \leq \pi$.

(b) (i) Using an appropriate double-angle formula, derive the identity for $\sin 2x$.

[2]

Recall:- Compound angle formula

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

When $B = A$,

$\sin(A + A) = \sin A \cos A + \cos A \sin A$

$\quad \sin 2A = 2 \sin A \cos A$

$\therefore \quad \sin 2x = 2 \sin x \cos x$

(ii) Hence, prove the identity $\frac{1 + \sin 2x}{\sin x + \cos x} = \sin x + \cos x$. [4]

L.H.S.

$$\frac{1 + \sin 2x}{\sin x + \cos x} = \frac{1 + \sin 2x}{\sin x + \cos x} \times \frac{\sin x + \cos x}{\sin x + \cos x}$$

$$= \frac{(1 + \sin 2x)(\sin x + \cos x)}{(\sin x + \cos x)(\sin x + \cos x)}$$

$$= \frac{(1 + \sin 2x)(\sin x + \cos x)}{\sin^2 x + \cos^2 x + 2 \sin x \cos x}$$

Recall: $\sin^2 x + \cos^2 x = 1$ and $\sin 2x = 2 \sin x \cos x$

$$= \frac{\cancel{(1 + \sin 2x)}(\sin x + \cos x)}{\cancel{(1 + \sin 2x)}}$$

$= \sin x + \cos x$

$=$ R.H.S.

Q.E.D.

(c) The position vectors of two points, A and B, whose coordinates are $(2, 3)$ and $(-3, 2)$ respectively are relative to an origin, O.

(i) Express the unit vector $\overrightarrow{AB}$ in terms of $x\boldsymbol{i} + y\boldsymbol{i}$. [4]

Since $A = (2, 3)$ then $\overrightarrow{OA} = \binom{2}{3}$

$$= 2\boldsymbol{i} + 3\boldsymbol{j}$$

Since $B = (-3, 2)$ then $\overrightarrow{OB} = \binom{-3}{2}$

$$= -3\boldsymbol{i} + 2\boldsymbol{j}$$

Using the triangle law,

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{0A}$$

$$= \binom{-3}{2} - \binom{2}{3}$$

$$= \binom{-5}{-1}$$

$$= -5\boldsymbol{i} - \boldsymbol{j}$$

$$|AB| = \sqrt{(-5)^2 + (-1)^2}$$

$$= \sqrt{25 + 1}$$

$$= \sqrt{26}$$

Substituting $\overrightarrow{AB} = -5\boldsymbol{i} - \boldsymbol{j}$ and $|AB| = \sqrt{26}$ into

$$Unit\ Vector\ of\ AB = \frac{\overrightarrow{AB}}{|AB|}$$

$$= \frac{-5\boldsymbol{i} - \boldsymbol{j}}{\sqrt{26}}$$

$$= \frac{1}{\sqrt{26}}(-5\boldsymbol{i} - \boldsymbol{j}) \quad or \quad -\frac{5}{\sqrt{26}}\boldsymbol{i} - \frac{1}{\sqrt{26}}\boldsymbol{j}$$

(ii) Find the angle $A\hat{O}B$, in degrees. [3]

$\overrightarrow{OA} = \binom{2}{3}$ $\overrightarrow{OB} = \binom{-3}{2}$

Using the dot product

$$\overrightarrow{OA}\,.\,\overrightarrow{OB} = (2 \times -3) + (3 \times 2)$$
$$= -6 + 6$$
$$= 0$$

$$|OA| = \sqrt{(2)^2 + (3)^2}$$
$$= \sqrt{4 + 9}$$
$$= \sqrt{13}$$

$$|OB| = \sqrt{(-3)^2 + (2)^2}$$
$$= \sqrt{9 + 3}$$
$$= \sqrt{13}$$

$$\cos\theta = \frac{\overrightarrow{OA}\,.\,\overrightarrow{OB}}{|OA||OB|}$$
$$\cos\theta = \frac{0}{\sqrt{13}\sqrt{13}}$$
$$\cos\theta = 0$$
$$\theta = \cos^{-1} 0$$
$$\theta = 90°$$

Therefore, angle $AOB = 90°$.

(iii) Hence, state the relationship between the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$. [1]

Since angle $AOB = 90°$, then the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are perpendicular to each other.

Total: 20 marks

SECTION III

INTRODUCTORY CALCULUS

ALL working must be clearly shown.

4. (a) Differentiate the following expression with respect to x, simplifying your answer.

$$y = \sin 4x + \frac{3}{x^4}$$

[3]

$$y = \sin 4x + \frac{3}{x^4}$$

$$y = \sin 4x + 3x^{-4}$$

$$\frac{dy}{dx} = 4\cos 4x + (-4)(3)x^{-4-1}$$

$$= 4\cos 4x - 12\,x^{-5}$$

$$= 4\cos 4x - \frac{12}{x^5}$$

(b) (i) Determine the coordinates of all the stationary points of the curve

$$y = \frac{1}{3}x^3 - \frac{5}{2}x^2 + 4x - \frac{1}{2}$$

[4]

$$y = \frac{1}{3}x^3 - \frac{5}{2}x^2 + 4x - \frac{1}{2}$$

$$= \frac{1}{3}x^3 - \frac{5}{2}x^2 + 4x^1 - \frac{1}{2}x^0$$

$$\frac{dy}{dx} = 3\left(\frac{1}{3}x^{3-1}\right) - 2\left(\frac{5}{2}x^{2-1}\right) + 1(4x^{1-1}) - 0\left(\frac{1}{2}x^{0-1}\right)$$

$$= x^2 - 5x + 4$$

At stationary points, $\frac{dy}{dx} = 0$.

$x^2 - 5x + 4 = 0$

$(x - 1)(x - 4) = 0$

Either

$x - 1 = 0$ or $x - 4 = 0$

$x = 1$ or $x = 4$

When $x = 1$,

$$y = \frac{1}{3}(1)^3 - \frac{5}{2}(1)^2 + 4(1) - \frac{1}{2}$$

$$y = \frac{1}{3} - \frac{5}{2} + 4 - \frac{1}{2}$$

$$y = \frac{4}{3}$$

When $x = 4$,

$$y = \frac{1}{3}(4)^3 - \frac{5}{2}(4)^2 + 4(4) - \frac{1}{2}$$

$$y = \frac{64}{3} - 40 + 16 - \frac{1}{2}$$

$$y = -\frac{19}{6}$$

$\therefore$ The stationary points are: $\left(1, \frac{4}{3}\right)$ and $\left(4, -\frac{19}{6}\right)$

(ii) Hence, determine the nature of EACH of the stationary points found in

(b) (i). [3]

$$\frac{dy}{dx} = x^2 - 5x + 4$$

$$= x^2 - 5x^1 + 4x^0$$

$$\frac{d^2y}{dx^2} = 2(x^{2-1}) - 1(5x^{1-1}) + 0(4x^{0-1})$$

$$= 2x - 5 + 0$$

$$= 2x - 5$$

When $x = 1$,

$$\frac{d^2y}{dx^2} = 2(1) - 5$$

$$= 2 - 5$$

$$= -3$$

Since $\frac{d^2y}{dx^2} < 0$, then the stationary point $\left(1, \frac{4}{3}\right)$ is a maximum point.

When $x = 4$,

$$\frac{d^2y}{dx^2} = 2(4) - 5$$

$$= 8 - 5$$

$$= 3$$

Since $\frac{d^2y}{dx^2} > 0$, then the stationary point $\left(4, -\frac{19}{6}\right)$ is a minimum point.

(c) The length of each edge of a cube is increasing at a rate of 0.25 $cm\ s^{-1}$.

Calculate the rate of change of the volume of the cube at the instant when the length is 8 cm. [5]

From the question, the following is given:

$$\frac{dx}{dt} = 0.25\ cm\ s^{-1}$$

$$Volume,\ \ v = x \times x \times x\ cm^3$$

$$= x^3\ cm^3$$

$$\frac{dv}{dx} = 3x^2$$

When $x = 8$,

$$\frac{dv}{dx} = 3(8)^2$$

$$= 192$$

Required to find: $\frac{dv}{dt}$

Using the chain rule,

$$\frac{dv}{dt} = \frac{dv}{dx} \times \frac{dx}{dt}$$

$$= 192 \times 0.25$$

$$= 48\ cm^3\ s^{-1}$$

$\therefore$ The rate of change of the volume is $\frac{dv}{dt} = 48\ cm^3\ s^{-1}$.

Total: 15 marks

5. (a) Determine

(i) $\int \sqrt{2x+1}\ dx$ [3]

$$\int \sqrt{2x+1}\ dx$$

$$= \int (2x+1)^{\frac{1}{2}}\ dx$$

$$= \frac{(2x+1)^{\frac{1}{2}+1}}{2\left(\frac{1}{2}+1\right)} + c$$

$$= \frac{(2x+1)^{\frac{3}{2}}}{3} + c$$

$$= \frac{1}{3}(2x+1)^{\frac{3}{2}} + c$$

$$\therefore \int \sqrt{2x+1}\ dx = \frac{1}{3}(2x+1)^{\frac{3}{2}} + c$$

STUDENT HUB

(ii) $\int \sin 2x + 5\cos x \; dx$ [3]

$$\int \sin 2x + 5\cos x \; dx$$

$$= \int \sin 2x + \int 5\cos x \, dx$$

Recall: $\int \sin ax \; dx = \frac{-\cos ax}{a} + c$

$$= \frac{-\cos 2x}{2} + 5\sin x + c$$

$$= -\frac{1}{2}\cos 2x \; + 5\sin x + c$$

(b) Figure 1 shows the region enclosed between the x-$axis$, the y-$axis$, the curve $y = x^2 + 2$ and the line $x = 3$.

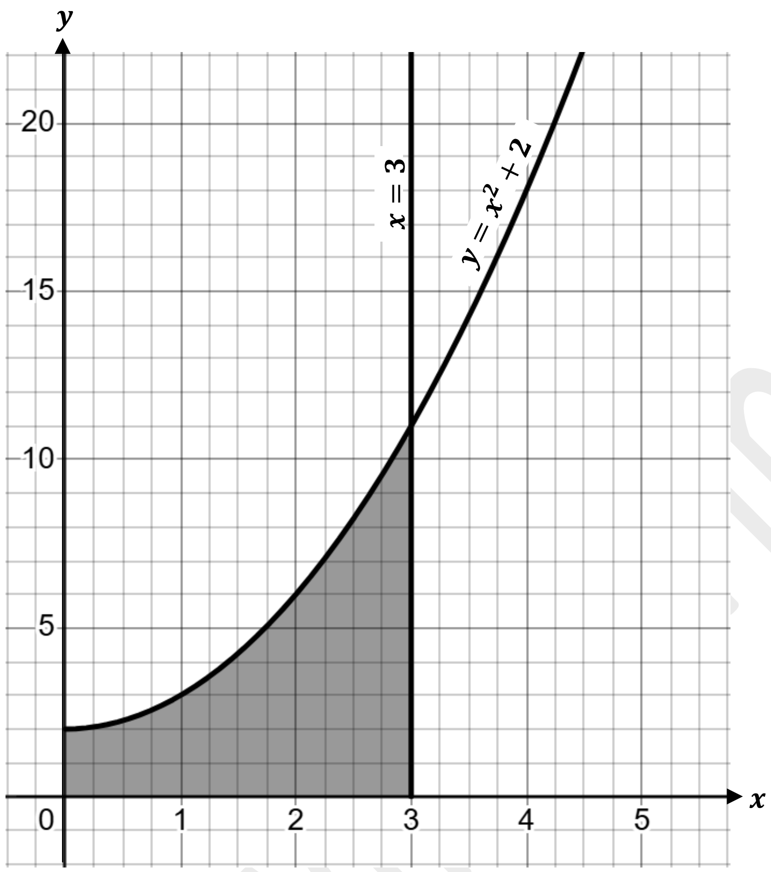


Figure 1. Region bounded by a curve, a line, the x-axis and y-axis.

Calculate the area of the shaded region. [4]

From the question, $y = x^2 + 2$, $a = 0$, $b = 3$.

$$\text{Area of shaded region, } A = \int_a^b y \ dx$$

$$= \int_0^3 x^2 + 2 \ dx$$

$$= \left[\frac{x^3}{3} + 2x\right]_0^3$$

$$= \left[\frac{(3)^3}{3} + 2(3)\right] - \left[\frac{(0)^3}{3} + 2(0)\right]$$

$$= 15\ units^2$$

(c) Figure 2 shows the shaded region enclosed between the x-axis, the y-axis and the curve $y = 4 - x^2$.

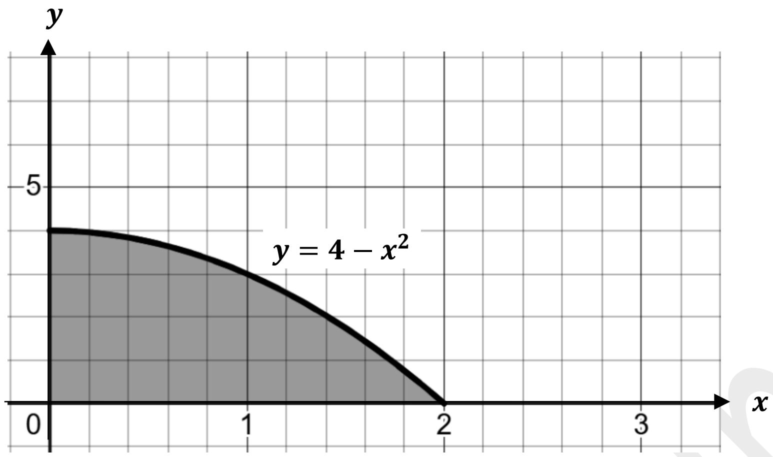


Figure 2. Region bounded by a curve, the x-axis, and y-axis.

If the region is rotated about the x-axis through one complete revolution, calculate the volume of the solid formed. [5]

From the question, $y = 4 - x^2, \quad a = 0, \quad b = 2.$

$$Volume,\ V = \pi \int_a^b y^2\ dx$$

$$= \pi \int_0^2 (4 - x^2)^2\ dx$$

$$= \pi \int_0^2 (16 - 8x^2 + x^4)\ dx$$

$$= \pi \int_0^2 (16x^0 - 8x^2 + x^4)\ dx$$

$$= \pi \left[\frac{16x^{0+1}}{1} - \frac{8x^{2+1}}{3} + \frac{x^{4+1}}{5}\right]_0^2$$

$$= \pi\left[16x - \frac{8x^3}{3} + \frac{x^5}{5}\right]_0^2$$

$$= \pi\left[\left(16(2) - \frac{8(2)^3}{3} + \frac{(2)^5}{5}\right) - \left(16(0) - \frac{8(0)^3}{3} + \frac{(0)^5}{5}\right)\right]$$

$$= \pi\left[\left(32 - \frac{64}{3} + \frac{32}{5}\right) - 0\right]$$

$$= \pi\left[\frac{256}{15}\right]$$

$$= \frac{256}{15}\pi \ \ units^3$$

Total: 15 marks

SECTION IV

PROBABILITY AND STATISTICS

ALL working must be clearly shown.

6. (a) If $P(A) = 0.8$, $P(B) = 0.5$ and $P(A \cup B) = 0.9$, calculate $P(B|A)$ [5]

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.8 + 0.5 - 0.9$

$= 0.4$

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$= \frac{0.4}{0.8}$$

$= 0.5$

$\therefore P(B|A) = 0.5$

(b) A fair, blue dice has each of the numbers 1, 3, 5, 7, 9 and 11 on its faces. A second, fair, red dice has each of the numbers 2, 4, 6, 8, 10 and 12 on its faces. The dice are rolled simultaneously and the sum of the numbers on their faces is recorded.

(i) Complete the possibility space table below by inserting the sum of the two dice on each roll. The first sum and the last sum have been inserted for you. [3]

	Red Dice						
	+	**2**	**4**	**6**	**8**	**10**	**12**
Blue Dice	**1**	3	5	7	9	11	13
	3	5	7	9	11	13	15
	5	7	9	11	13	15	17
	7	9	11	13	15	17	19
	9	11	13	15	17	19	21
	11	13	15	17	19	21	23

(ii) State the probability that the sum of the scores from throwing the dice at (b)(i) is

- Prime [1]

From the possibility space table, the prime numbers are:

2, 3, 5, 7, 11, 13, 17, 19, 23.

In total, there were 25 prime numbers out of a total of 36 outcomes.

$$P(\textit{the sum is prime}) = \frac{\textit{Number of prime sums}}{\textit{Total number of sums}}$$

$$= \frac{25}{36}$$

$\therefore$ The probability that the sum of the scores is prime is $\frac{25}{36}$.

- Even [1]

There are no even numbers in the possibility space table.

$$P(\textit{the sum is even}) = \frac{\textit{Number of even sums}}{\textit{Total number of sums}}$$

$$= \frac{0}{36}$$

$$= 0$$

$\therefore$ The probability that the sum of the scores is even is 0.

(c) Calculate the interquartile range of the data set represented in the stem-and-leaf diagram below. [3]

Stem	Leaf					
3	0	1	3	3		
4	2	4	5			
5	1	6	6	8	9	9
6	4	7	7			
7	3	6				
8	5					
9	2					

Key: 3|0 means 30

The data set has 20 values in total so $n = 20$.

$$Q_1 = \frac{1}{4}(n+1)^{th} \text{ value}$$
$$= \frac{1}{4}(20+1)^{th}$$
$$= \frac{1}{4}(21)^{th}$$
$$= 5.25^{th} \text{ value}$$

Looking at the stem-and-leaf diagram, the 5th value is 42 and the 6th value is 44.

Therefore,

$$Q_1 = \frac{42+44}{2}$$
$$= 43$$

$Q_3 = \frac{3}{4}(n+1)^{th}$ value

$= \frac{3}{4}(20+1)^{th}$

$= \frac{3}{4}(21)^{th}$

$= 15.75^{th}$ value

Looking at the stem-and-leaf diagram, the 15th value is 67 and the 16th value is 67.

Therefore,

$$Q_3 = \frac{67+67}{2}$$

$$= \frac{134}{2}$$

$$= 67$$

Interquartile range, IQR $= Q_3 - Q_1$

$= 67 - 43$

$= 24$

$\therefore$ The interquartile range is 24.

(d) Figure 3 shows a box-and-whisker plot which represents a set of numerical data.

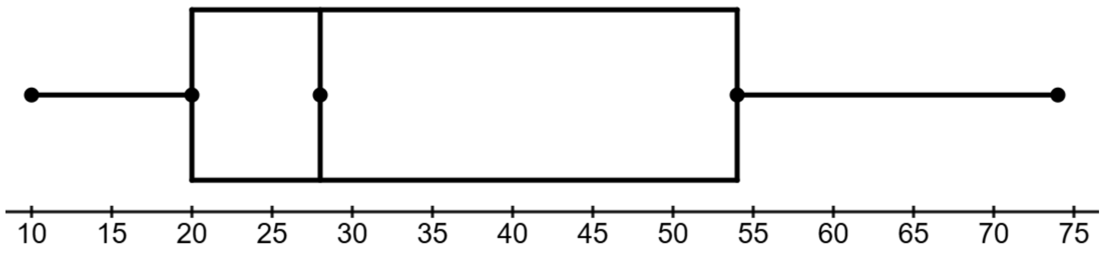


Figure 3. Box-and-whisker plot

(i) State the median of the data set. [1]

Reading off the box-and-whisker plot,

Median $= 28$.

(ii) Determine the range of the data set. [1]

Range $=$ Highest value $-$ Lowest value

$= 74 - 10$

$= 64$

(iii) Describe the shape of the distribution. [1]

Since Q_2 is closer to Q_1, the distribution is positively skewed and since the right whisker is longer, we can say that there is more variation in the upper half of the data.

(e) Five out of 30 students in a class were randomly selected and their heights recorded in centimetres as shown below.

Student	Height
A	135
B	144
C	137
D	146
E	153

Calculate the variance of their heights, in square centimetres. [4]

$$\text{The mean height, } \bar{x} = \frac{\sum x}{n}$$

$$= \frac{715}{5}$$

$$= 143$$

Student	**Height, x**	$(x - \bar{x})^2$
A	135	64
B	144	1
C	137	36
D	146	9
E	153	100

$$\sum (x - \bar{x})^2 = 210$$

Variance, $s = \frac{\sum(x-\bar{x})^2}{n-1}$

$= \frac{210}{5-1}$

$= \frac{210}{4}$

$= 52.5\ cm^2$

Total: 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.